

# The Arbitrage-Free Valuation Roadmap: CFA Level II Fixed Income

## 1. Foundations of No-Arbitrage

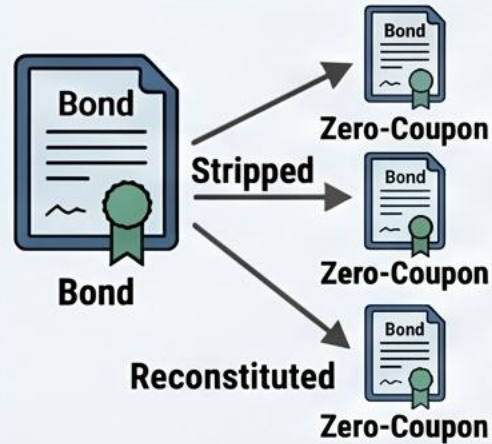


**Arbitrage-Free Valuation:** Determining the value consistent with the absence of riskless profit.

**Law of One Price:** Perfect substitutes must sell for the same price.

**Value Additivity:** Value of the part.

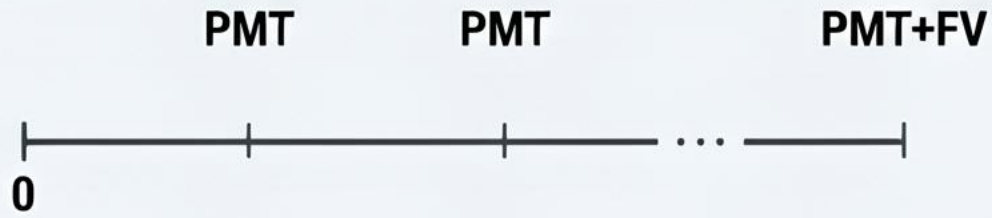
**Stripping & Reconstitution** exploits price discrepancies.



### Arbitrage Opportunity Example

| Asset   | Price Today | Payoff in 1 Year | Arbitrage Cehistation                                                              |
|---------|-------------|------------------|------------------------------------------------------------------------------------|
| Asset A | 0.952381    | \$1.00           | \$1 / 1.05 = 0.952281 (Fair Value)                                                 |
| Asset B | \$97.00     | \$105.00         | Opportunity: \$105 x 0.952381 = 100<br>Buy B at \$97, Sell 105 of A to profit \$5. |

## 2. Valuing Option-Free Bonds with Spot Rates



$$PV = \frac{PMT}{(1 + Z_1)^1} + \frac{PMT}{(1 + Z_2)^2} + \dots + \frac{PMT + FV}{(1 + Z_n)^n}$$

**Bond as a Package of Zeros:** Discounting each cash flow at its specific spot rate.

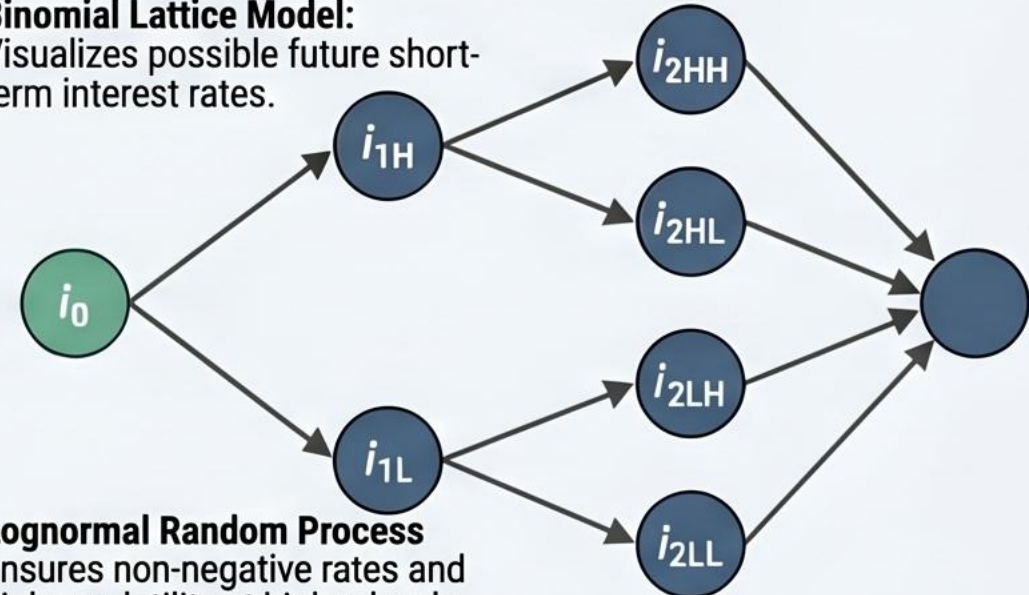
**Spot Rate ( $Z_n$ ):** Market discount rate for a single payment at maturity N, often derived from Treasury yields using bootstrapping.

Spot rates are more precise than YTM, capturing term structure variation.



## 3. The Binomial Interest Rate Tree Framework

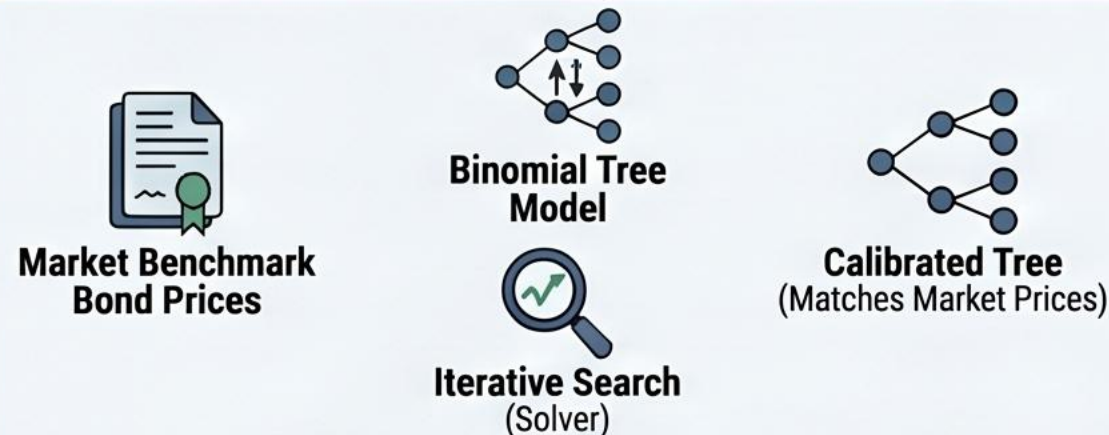
**Binomial Lattice Model:** Visualizes possible future short-term interest rates.



**Lognormal Random Process** ensures non-negative rates and higher volatility at higher levels.

**Node Relationship:** Adjacent rates related by  $R_u = R_d \cdot e^{2\sigma}$  (where  $\sigma$ )  
**Recombining Trees** improve computational efficiency.

## 4. Calibration to the Term Structure

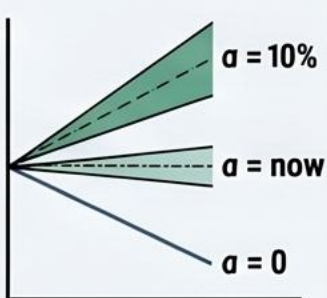


**Calibration:** Fitting the tree nodes so model prices match observed market benchmark bond prices.

**Iterative Search:** Finds forward rates that produce correct values.

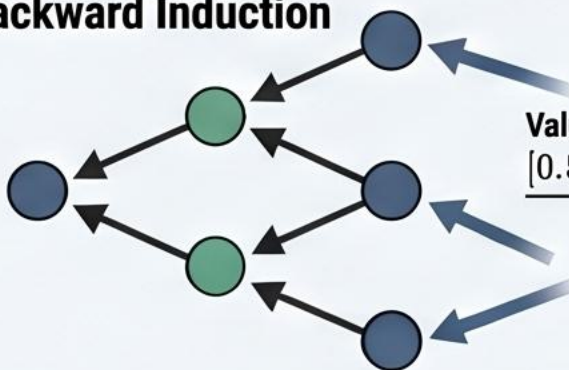
**Volatility Impact:** Higher volatility ( $\sigma$ ) widens the spread between  $R_0$  and  $R_d$ .

**Arbitrage-Free Check:** The calibrated tree must correctly price underlying par bonds.



## 5. Advanced Valuation Methodologies

### Backward Induction

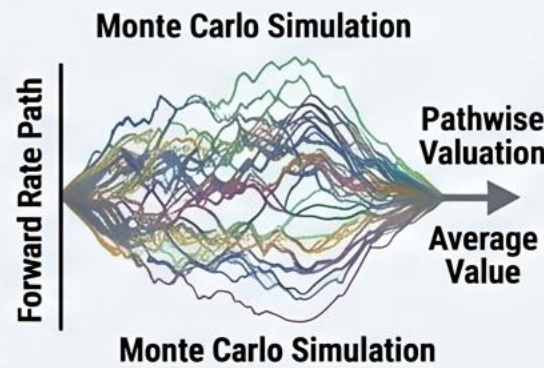


$$\text{Value at Node} = \frac{[0.5 \times (V_u + C) + 0.5 \times (V_d + C)]}{(1 + r)}$$

**Backward Induction:** Valuing a bond by starting at maturity and working back to Time 0.

**Pathwise Valuation:** Calculating value for every possible path and averaging.

**Monte Carlo Simulation:** Used for path-dependent cash flows, simulating drift-adjusted rate paths.



## 6. Term Structure Models & Exam Traps

### CIR vs. Vasicek Models

| Model   | Volatility Assumption     | Key Feature    |
|---------|---------------------------|----------------|
| CIR     | Increases with rate level | Mean-Reverting |
| Vasicek | Constant                  | Mean-Reverting |

### Mean Reversion



**Drift Term:** Expected change in interest rates over time.

**Exam Trap:** Spot rates fail for bonds with embedded options as cash flows are interest-rate dependent, lattice models are required.